

Mathematical Statistics

Lecture 08 The Bernoulli Distribution

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1 Introduction to the Bernoulli Distribution

This lecture introduces the most fundamental discrete probability distribution: the **Bernoulli Distribution**. It is the starting point for understanding more complex distributions and is crucial for many applications in statistics and machine learning.

Learning Objectives

By the end of this lecture, you will be able to:

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- Define the Bernoulli distribution and its key characteristics.
- Identify real-world scenarios that can be modeled as Bernoulli trials.
- Understand and use the Probability Mass Function (PMF).
- Calculate the Expected Value and Variance of a Bernoulli random variable.

2 Defining the Bernoulli Trial

A Bernoulli distribution is a **discrete probability distribution** that models the outcome of a single experiment, known as a **Bernoulli trial**.

2.1 Key Characteristics

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- **Single Trial:** The experiment is performed only once.
- **Two Possible Outcomes:** The trial can only result in one of two outcomes. By convention, these are labeled:
 - **Success:** Assigned the numerical value **1**.
 - **Failure:** Assigned the numerical value **0**.

2.2 Illustrative Examples

2.2.1 Example 1: Tossing a Fair Coin

A single flip of a coin is a classic Bernoulli trial.

- **Random Variable (X):** $X = 1$ for Heads (Success), $X = 0$ for Tails (Failure).

2.2.2 Example 2: Rolling a Single Die

This example shows how we can frame a multi-outcome experiment as a Bernoulli trial by defining our event of interest.

- **Experiment:** Roll a standard six-sided die once.
- **Bernoulli Framing:** Let's define "success" as rolling a 6.
 - **Success ($X = 1$):** The outcome is a 6.
 - **Failure ($X = 0$):** The outcome is any other number {1, 2, 3, 4, 5}.

- Even with six possible results, we have created a binary "yes/no" scenario: "Was it a 6?". This is a Bernoulli trial.

3 Formal Representation

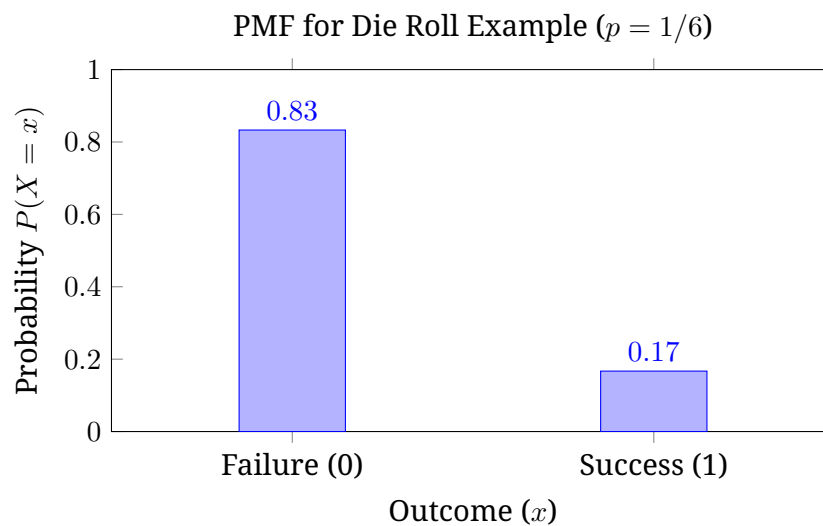
3.1 Probability Distribution Tables

A distribution table shows the random variable's values and their corresponding probabilities, denoted $P(X)$.

Table 1: Die Roll Example (Success = rolling a 6)		
Random Variable (X)	Outcome(s)	Probability $P(X)$
0	{1, 2, 3, 4, 5}	$5/6 \approx 0.833$
1	{6}	$1/6 \approx 0.167$

3.2 Graphical Representation

We can visualize this discrete distribution with a bar chart.



4 Mathematical Foundations

💡 Core Formulas

A Bernoulli distribution is defined by a single parameter, p , the probability of success.

- $P(X = 1) = p$
- $P(X = 0) = 1 - p$

Probability Mass Function (PMF)

The PMF gives the probability of any outcome $x \in \{0, 1\}$ in a single, compact formula:

$$f(x) = P(X = x) = p^x(1 - p)^{1-x}$$

Verification:

- For Success ($x = 1$): $f(1) = p^1(1 - p)^{1-1} = p \cdot (1 - p)^0 = p$.
- For Failure ($x = 0$): $f(0) = p^0(1 - p)^{1-0} = 1 \cdot (1 - p)^1 = 1 - p$.

Expected Value (Mean)

The expected value, $E[X]$, is the long-term average outcome. It is calculated as $E[X] = \sum x \cdot P(X = x)$.

$$\begin{aligned} E[X] &= (0 \cdot P(X = 0)) + (1 \cdot P(X = 1)) \\ &= (0 \cdot (1 - p)) + (1 \cdot p) \\ &= 0 + p = p \end{aligned}$$

The expected value is simply the probability of success, $E[X] = p$.

Variance

The variance, $\text{Var}(X)$, measures the spread of the distribution, calculated as $\text{Var}(X) = E[X^2] - (E[X])^2$.

1. **First, find $E[X^2]$:**

$$\begin{aligned} E[X^2] &= \sum x^2 \cdot P(X = x) \\ &= (0^2 \cdot P(X = 0)) + (1^2 \cdot P(X = 1)) \\ &= (0 \cdot (1 - p)) + (1 \cdot p) = p \end{aligned}$$

2. **Now, calculate the variance:**

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= p - (p)^2 = p - p^2 \end{aligned}$$

Factoring out p , the variance is $\text{Var}(X) = p(1 - p)$.

Practical Applications in Data Science & Machine Learning

The Bernoulli distribution is the foundation for modeling many binary events:

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- **A/B Testing:** Modeling user clicks on different website designs (Click=1, No Click=0).
- **Binary Classification:** The output of models like Logistic Regression is the probability p of an item belonging to a class. The outcome (e.g., Spam=1, Not Spam=0) is a Bernoulli variable.
- **Churn Prediction:** Modeling whether a customer will cancel their subscription (Churn=1, No Churn=0).
- **Fraud Detection:** Modeling if a transaction is fraudulent (Fraud=1, Not Fraud=0).

5 Practice Problems

1. A factory produces light bulbs, and 5% are defective. A single bulb is tested. Let $X = 1$ if the bulb is defective and $X = 0$ otherwise.
 - (a) What is the value of the parameter p ?
 - (b) What is the expected value, $E[X]$?
 - (c) What is the variance, $\text{Var}(X)$?
2. You play a game where you win by drawing an Ace from a standard 52-card deck.
 - (a) Define the random variable X , the success/failure outcomes, and the parameter p .
 - (b) What is the probability of failure?
3. A Bernoulli random variable X has an expected value of 0.3.
 - (a) What is $P(X = 1)$?
 - (b) What is $P(X = 0)$?
 - (c) What is the variance of X ?

6 Solutions to Practice Problems

1. Light Bulb Problem:

- (a) Success is a defective bulb. The probability is 5%, so $p = 0.05$.
- (b) $E[X] = p = 0.05$.
- (c) $\text{Var}(X) = p(1 - p) = 0.05(1 - 0.05) = 0.05(0.95) = 0.0475$.

2. Card Game Problem:

- (a) X is the outcome of the draw. Success ($X = 1$) is drawing an Ace. Failure ($X = 0$) is drawing any other card. There are 4 Aces in 52 cards, so $p = \frac{4}{52} = \frac{1}{13}$.
- (b) Probability of failure is $1 - p = 1 - \frac{1}{13} = \frac{12}{13}$.

3. Expected Value Problem:

- (a) We know $E[X] = p$ and $P(X = 1) = p$. Since $E[X] = 0.3$, then $P(X = 1) = 0.3$.
- (b) $P(X = 0) = 1 - p = 1 - 0.3 = 0.7$.
- (c) $\text{Var}(X) = p(1 - p) = 0.3(1 - 0.3) = 0.3(0.7) = 0.21$.